

Peer Response

Hi Milosz, your analysis provides a clear and well-structured comparison between VeraCrypt and BitLocker, particularly through the Weakness → Trigger → Impact model, which effectively illustrates causal reasoning. This aligns well with principles of secure software development, where identifying underlying causes is central rather than focusing on isolated vulnerabilities.

A strong point in your argument is the recognition that the security of VeraCrypt depends on user behaviour. This reflects a broader issue within cybersecurity, where human factors often undermine technically robust systems. As emphasised in the literature on secure development, security measures must account for realistic user behaviour rather than ideal conditions (Olmsted, 2024). In software development, these 'ideal conditions' are often referred to as the 'happy flow'. Your discussion of weak passwords and improper keyfile management clearly illustrates organisational and user behaviour risks.

However, your classification of "lack of hardware integration" as a vulnerability with medium severity could be further refined. While this increases reliance on user-managed secrets, it is also a deliberate design choice that supports platform independence. In contrast, BitLocker's reliance on TPM introduces a different risk profile, including potential vulnerabilities at the hardware or firmware level, which you might explore further.

Additionally, your point regarding the absence of recovery mechanisms is valid, but it may be useful to draw a clearer distinction between security and availability, as captured in the well-known CIA triad. The absence of recovery options strengthens confidentiality but has direct implications for availability, highlighting the inherent trade-offs within security design.

Overall, your contribution demonstrates strong analytical depth and a solid integration of both technical and human-centred perspectives.

Reference

Olmsted, A. (2024) *Security-Driven Software Development*. Birmingham: Packt Publishing.